

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	05-07-2026
Team ID	SWTID-2026-8108
Project Name	Heart Disease Analysis
Maximum Marks	

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Authentication & Access Control	* Secure login interface for Clinicians and Analysts via database credentials. * Role-based routing to distinct clinical or public policy view panes.
FR-2	Interactive Clinical Cross-Filtering	* Filter dashboard worksheets concurrently by Age, Sex, and Smoking Status. * Real-time dashboard update upon changing patient cohort filters.
FR-3	Demographic & Regional Segments	* View multi-tab demographic splits comparing Urban vs. Rural populations. * Display disease prevalence bar charts filtered by racial backgrounds.
FR-4	Live Aggregate Profiling	* Display live average BMI shifts across dynamic diabetic status bands. * Show physical activity metrics mapped against diagnosis counts
FR-5	Personal Progress Tracking	* Display color-coded benchmark comparison gauges for single patient metrics. * Visual status warning cards showing high-risk vs. healthy zones

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	The solution must utilize an integrated, tiled Tableau Storyboard layout to ensure clean, easy-to-read scannability on a single screen without messy, isolated worksheets
NFR-2	Security	Database access must require local MySQL account validation to prevent unauthorized updates or exposure of clinical health metrics.
NFR-3	Reliability	Dashboard visuals must precisely map data points directly from the database schema without dropouts or visual corruption across all 18 data columns
NFR-4	Performance	Heavy analytical aggregate queries across the entire 4,416-row dataset must execute and update the dashboard layout smoothly in under 3 seconds using the MySQL
NFR-5	Availability	The local MySQL relational server system and attached Tableau reporting interface must be accessible at all times during presentation run cycles.
NFR-6	Scalability	The database and dashboard query pipelines must be architected to smoothly handle updates if more patient metric rows are appended to the main table.